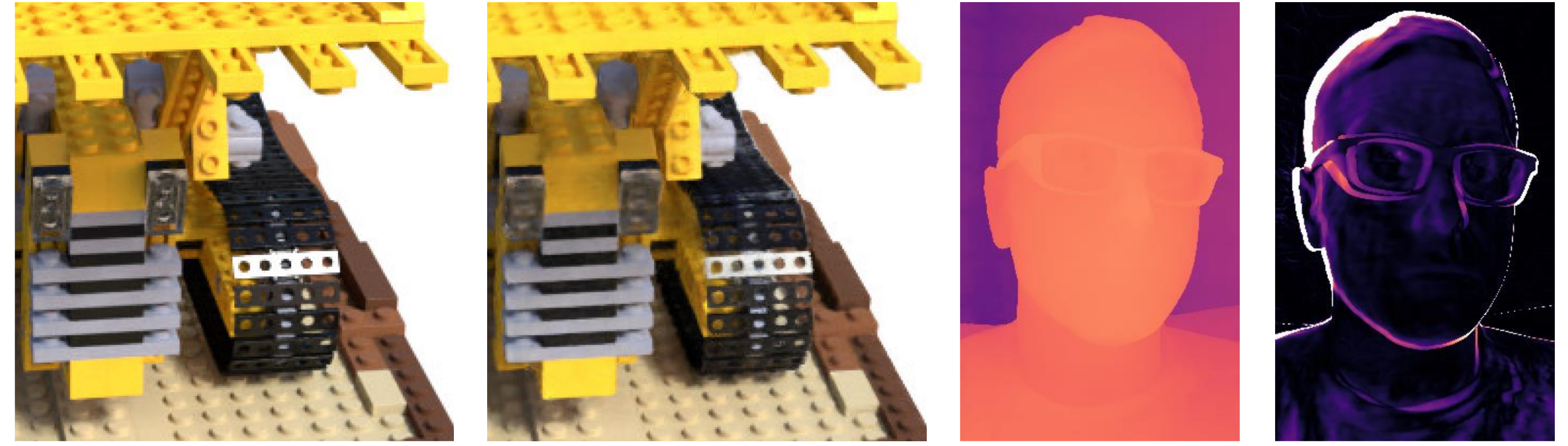


Error Measurement and Visualization in 3D Volume Data

Background

Numerous approaches have been explored for reconstructing 3D space from multiple images. Recent advancements in deep learning techniques have sparked renewed interest in volume based reconstruction methods, exemplified by the emergence of Neural Radiance Field (NeRF). Due to its ability to generate realistic virtual views of complex objects using simple network architectures, NeRF has been extensively investigated in subsequent research. Most studies evaluating reconstructed results rely on peak signal-to-noise ratio (PSNR) and structural similarity index map (SSIM) as metrics for error assessment. However, they suffer from the drawback of amplifying errors in the viewpoint when shifts occur in the x and y directions rather than the depth direction. Consequently, the difference in error values can vary depending on the viewpoint.



PSNR 40.15, SSIM 0.991

- [1] B. Mildenhall, *et al.*, "NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis," ECCV 2020.
[2] K. Park, *et al.*, "Nerfies: Deformable Neural Radiance Fields," ICCV 2021.

We presents a novel method for quantitatively measuring and visually comparing errors which calculates the error for each pixel by computing its distance to the nearest neighbor in 3D space. The error measurement becomes viewpoint-independent, preventing significant variations in error values based on the viewpoint. Moreover, the method enables meaningful visualization of errors in both the x and y directions, as well as the depth direction.

Proposed Method

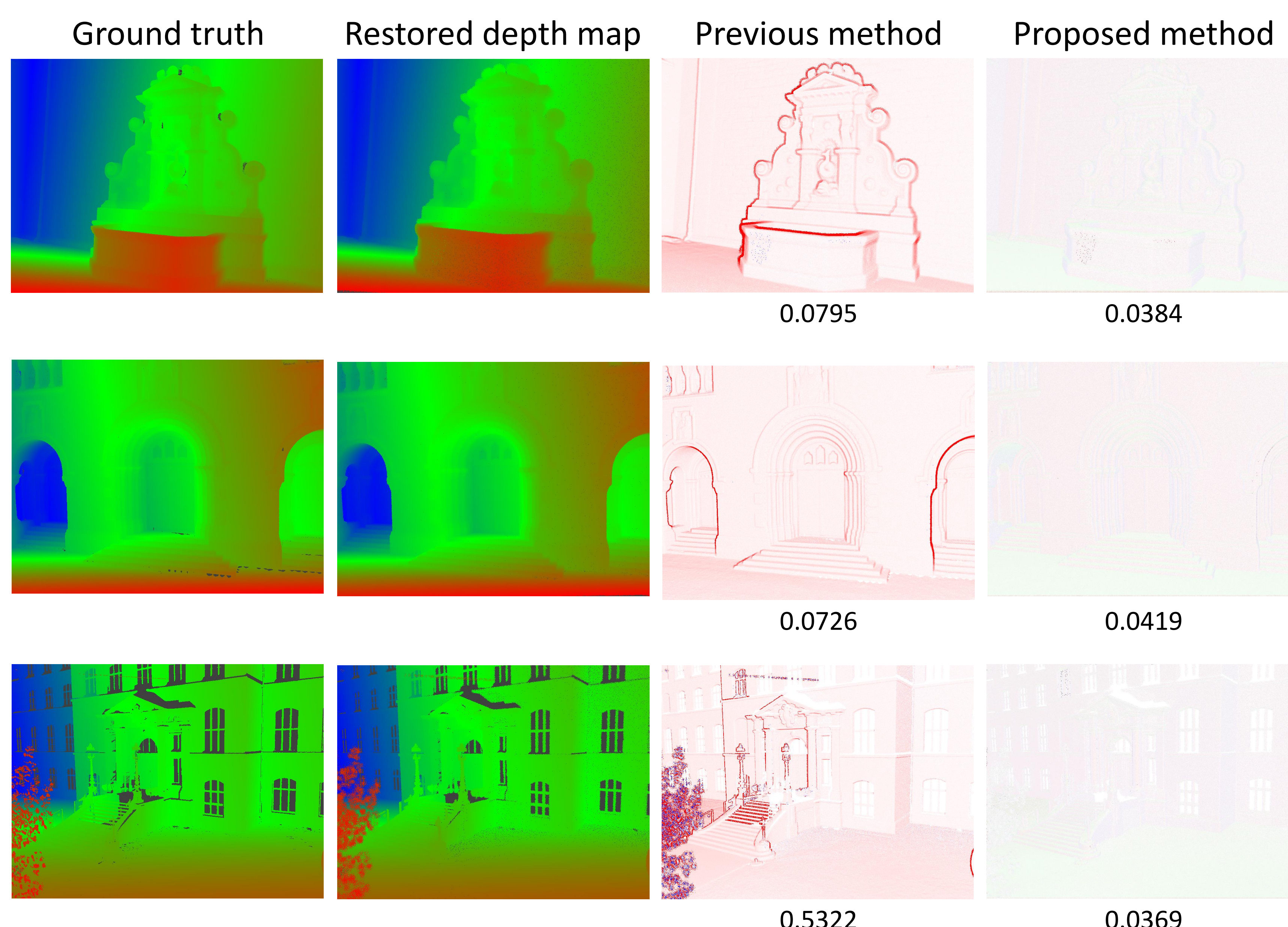
Algorithm 1 provides a description of the process for computing and visualizing the error based on two input depth maps.

We generate depth images D_s and D_t by volumetrically rendering each model from the camera's viewpoint. Next, we convert each depth image D_s and D_t into 3D point clouds P_s and P_t respectively, using the intrinsic parameter K . For each vertex v_i in the source point cloud P_s , we compare its distance to every vertex v_j in the target point cloud P_t . The error of vertex v_i is determined by storing the distance to its closest neighboring vertex in P_t . To determine the color value of the pixel corresponding to v_i in the depth map D_s , we utilize the distances in the x, y, and z directions between vertex v_i and the closest vertex v_j in P_t . These distances are used to compute the RGB color representation. Finally, the error is computed as the sum of all individual errors e_i . To visually represent these errors, an error image can be constructed as a set of color values c_i associated with each pixel.

Algorithm 1: Measure and visualize errors in two depth images

Data: source depth map D_s , target depth map D_t , intrinsic parameter K , scale s
Result: error value E , error image I_E
Generate 3D point cloud P_s and P_t from D_s and D_t using K ;
foreach $v_i \in P_s$ **do**
 foreach $v_j \in P_t$ **do**
 Calculate Euclidean distance d_{ij} between v_i and v_j ;
 $\mathcal{D}_i \leftarrow \mathcal{D}_i \cup d_{ij}$;
 end
 $e_i = \min_{d_{ji} \in \mathcal{D}_i} d_{ij}$;
 $c_i \leftarrow (255 - \lfloor x(e_i)/s \rfloor, 255 - \lfloor y(e_i)/s \rfloor, 255 - \lfloor z(e_i)/s \rfloor)$;
 $E = E + e_i$;
 $I_E \leftarrow I_E \cup c_i$;
end
return E, I_E

Results



Conclusion

We have presented novel methods for measuring and visualizing errors in 3D volume data. Our approaches address the limitations of existing techniques by offering reliable and consistent error measurements, independent of the camera's viewpoint. The visualization method accurately represents errors in all directions, facilitating a comprehensive analysis of reconstructed 3D volume data. The experimental evaluations validate the effectiveness of our proposed methods and highlight their potential for improving error analysis in various applications involving 3D volume data.